



Advanced Time Series Analysis and Forecasting

Seminar 15: Review and project defence



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ **ACI**: Adaptive Conformal Inference
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CW**: Clark–West (test for nested models)
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **FWER**: Family-Wise Error Rate
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP**: Gross Domestic Product
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MA**: Moving Average
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **RMSPE**: Root Mean Squared Prediction Error
- ▣ **SVAR**: Structural Vector AutoRegression
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VECM**: Vector Error Correction Model

Today's question and seminar structure (1/4)

- The question of the seminar
 - ▶ **can you use the tools of the whole course quickly and defend a replication result under questioning?**
- The seminar is a project clinic on [Chapters 0–14](#)
- Three types of exercises
 - ▶ **A**: short problems from [Chapters 0–14](#), on paper, of the kind asked at the defence
 - ▶ **B**: mock defences on replication outputs, in the notebook
 - ▶ **C**: the pre-registration of your own project and the correction of an answer given by an AI tool
- Each exercise is marked
 - ▶ **[Solved]**: the full solution is presented in the seminar, in the slides and in the notebook; it is the model to follow
 - ▶ **[Proposed]**: you solve it yourselves, following the model; the solution is discussed in the seminar

Today's question and seminar structure (2/4)

- ▣ Part A: short problems across the course
 - ▶ A1: how much does a mean know? (Chapter 0)
 - ▶ A2: a Diebold–Mariano test by hand (Chapter 1)
 - ▶ A3: Cholesky responses (Chapter 3)
 - ▶ A4: the half-life of an equilibrium error (Chapter 4)
 - ▶ A5: one step of the Kalman filter (Chapter 6)
 - ▶ A6: a two-state Markov chain (Chapter 7)
 - ▶ A7: the Kupiec test for VaR 1% (Chapter 9)
 - ▶ A8: GARCH persistence (Chapter 8)
 - ▶ A9: the split-conformal quantile (Chapter 13)
 - ▶ A10: placebo p-values and Holm (Chapters 14 and 0)

Today's question and seminar structure (3/4)

- ▣ Part B: mock defences on replication outputs
 - ▶ B1: Meese and Rogoff for EUR/RON
 - ▶ B2: the term spread and growth
 - ▶ B3: a leakage audit
 - ▶ B4: a rule search on the BET
- ▣ Part C: your project and an AI critique
 - ▶ C1: the pre-registration of your project
 - ▶ C2: audit an AI answer

Today's question and seminar structure (4/4)

- Outputs to produce
 - ▶ the answers to the A exercises
 - ▶ for each B exercise, the answers you would give at the defence, with the numbers that support them
 - ▶ the pre-registration of your project (C1) and the verdicts on the AI answer in C2, each with one line of justification
 - ▶ one sentence of interpretation for each exercise
- Notebook for today: [open the seminar notebook in Google Colab](#)
 - ▶ each task names its notebook section
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the long-run variance of a mean (Ch. 0); a DM test by hand (Ch. 1)	Solved, Proposed	A1
A3, A4	Cholesky responses (Ch. 3); the half-life of an equilibrium error (Ch. 4)	Solved, Proposed	A3
A5, A6	one Kalman step (Ch. 6); a two-state Markov chain (Ch. 7)	Solved, Proposed	A5
A7, A8	the Kupiec test (Ch. 9); GARCH persistence (Ch. 8)	Solved, Proposed	A7
A9, A10	the split-conformal quantile (Ch. 13); placebo p-values and Holm (Ch. 14, 0)	Solved, Proposed	A9
B1, B2	mock defences: Meese–Rogoff for EUR/RON; the term spread	Solved, Proposed	B1
B3, B4	mock defences: a leakage audit; a rule search on the BET	Solved, Proposed	B3
C1, C2	your pre-registration with a power calculation; what is wrong in an AI answer?	Proposed	B1, A1–A10

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
EUR/RON reference rate	BNR	monthly (end of month)	2005–2026
US real GDP, 10-year and 3-month Treasury yields	FRED	quarterly	1962–2025
BET index	EODHD	daily	2001–2012
Simulated data (leakage audit)	–	–	–

- In the notebook: `dm_test`, `lr_v`, `ols_se`, `leakage_r2`, `stationary_bootstrap_index`, `p_required`; no account or key is needed

Prerequisites for Today (1/20): the long-run variance

- ▣ The **long-run variance** sets the precision of a mean of dependent data: $\text{Var}(\bar{x}) \approx \Omega/T$

$$\Omega = \gamma_0 + 2 \sum_{j \geq 1} \gamma_j$$

- ▶ γ_j : autocovariance at lag j (γ_0 : the variance); T : sample size

- ▣ **AR(1)** $x_t = \phi x_{t-1} + \varepsilon_t$, innovation variance σ^2

$$\gamma_0 = \frac{\sigma^2}{1 - \phi^2}, \quad \Omega = \frac{\sigma^2}{(1 - \phi)^2}, \quad \frac{\Omega}{\gamma_0} = \frac{1 + \phi}{1 - \phi}$$

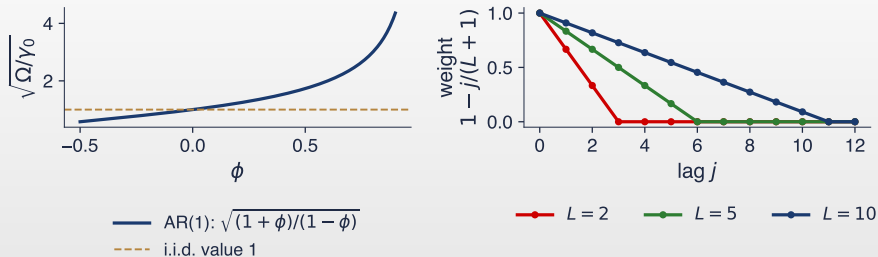
- ▶ $\sqrt{\Omega/\gamma_0}$: the factor by which i.i.d. standard errors are too small

- ▣ **Newey–West** estimator [Newey and West \(1987\)](#): weighted sum of sample autocovariances

$$\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) \hat{\gamma}_j, \quad \hat{\gamma}_j = \frac{1}{T} \sum_{t>j} (x_t - \bar{x})(x_{t-j} - \bar{x})$$

- ▶ L : number of lags (rule $L = \lfloor 4(T/100)^{2/9} \rfloor$, $\lfloor \cdot \rfloor$: integer part); $1 - j/(L+1)$: Bartlett weights

Prerequisites for Today (2/20): dependence and the precision of a mean



- Left: the factor $\sqrt{\Omega/\gamma_0}$ by which i.i.d. standard errors must be multiplied for an AR(1)
 - ▶ 1.36 for $\phi = 0.3$, 4.36 for $\phi = 0.9$; below 1 for $\phi < 0$ (negative autocorrelation)
- Right: the Bartlett weights $1 - j/(L+1)$ of $\hat{\gamma}_j$ in $\hat{\Omega}$; they decline linearly and vanish after lag L
 - ▶ the rule $L = \lfloor 4(T/100)^{2/9} \rfloor$ gives $L = 4$ for $T = 250$ and $L = 6$ for $T = 1000$; the weights keep $\hat{\Omega} \geq 0$

Prerequisites for Today (3/20): comparing two forecasts

- **DM test** Diebold and Mariano (1995): is the expected loss of forecast 1 equal to that of forecast 2?

$$d_t = L(e_{1t}) - L(e_{2t}), \quad \text{DM} = \bar{d} / \sqrt{\hat{\Omega}_d / P}$$

- ▶ e_{it} : forecast error of model i ; L : loss (e.g. squared error); $\bar{d}$: mean of d_t ; $\hat{\Omega}_d$: its long-run variance; P : number of forecasts

- **HLN correction** Harvey, Leybourne and Newbold (1997) for small samples and horizon h

$$\text{DM}^* = \text{DM} \sqrt{(P + 1 - 2h + h(h - 1)/P) / P}$$

- ▶ compared with Student- t with $P - 1$ degrees of freedom (t_{P-1})

- **Clark–West** Clark and West (2007) for nested models (model 1 is the smaller one)

$$f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$$

- ▶ $\hat{y}_{it}$: forecast of model i ; the bracket removes the noise of estimating the extra coefficients; one-sided test of $\mathbb{E}f_t \leq 0$

Prerequisites for Today (4/20): accuracy measures and nested models

- **RMSE** of P forecasts and the ratio to a benchmark

$$\text{RMSE} = \sqrt{\frac{1}{P} \sum_{t=1}^P e_t^2}, \quad \text{ratio} = \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{benchmark}}}$$

- ▶ ratio < 1: the model is more accurate; random walk for a log change: $\hat{y}_{t+1} = 0$

- **Clark–West** Clark and West (2007) statistic: the mean of the adjusted differential f_t over its standard error

$$\text{CW} = \frac{\bar{f}}{\sqrt{\hat{\Omega}_f/P}}, \quad \bar{f} = \frac{1}{P} \sum_{t=1}^P f_t$$

- ▶ $\hat{\Omega}_f$: the long-run variance of f_t ; reject the smaller model if $\text{CW} > 1.645$ (one-sided, 5%)

- **Out-of-sample R^2** : positive when the forecast beats the training mean $\bar{y}_{\text{train}}$

$$R_{\text{OOS}}^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y}_{\text{train}})^2}$$

- **t-statistic** of a slope: $t = \hat{\beta}/\text{se}(\hat{\beta})$, with a classical or a HAC standard error (Newey–West Newey and West (1987), L lags)

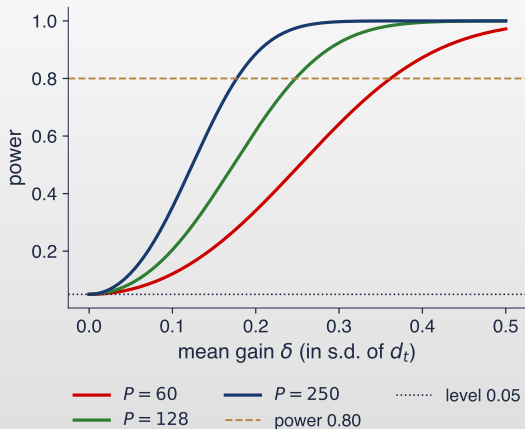
Prerequisites for Today (5/20): the power of a DM test

- Number of forecasts P needed to detect a mean gain δ with power π

$$P \approx \Omega_d \frac{(z_{1-a/2} + z_\pi)^2}{\delta^2}$$

- ▶ δ : the mean gain, in standard deviations of d_t ; a : the test level (e.g. 5%); z_q : the q -quantile of the standard Normal ($z_{0.975} = 1.96$, $z_{0.8} = 0.84$)
- ▶ Ω_d : long-run variance of the standardised d_t ; for an AR(1) differential with coefficient ρ , $\Omega_d = (1 + \rho)/(1 - \rho)$
- Reading
 - ▶ halving δ quadruples P ; positive dependence ($\rho > 0$) multiplies P by Ω_d
 - ▶ a non-rejection with P far below this value says little about the absence of a gain

Prerequisites for Today (6/20): power curves of the DM test



- Power of a two-sided 5% test, i.i.d. differential ($\Omega_d = 1$)

$$\pi(\delta) = \Phi(\delta\sqrt{P} - 1.96) + \Phi(-\delta\sqrt{P} - 1.96)$$

- ▶ Φ : the standard Normal distribution function; δ : the mean gain in standard deviations of d_t

- For $\delta = 0.2$: power 0.34 with $P = 60$, 0.62 with $P = 128$, 0.89 with $P = 250$

- ▶ at $\delta = 0$ every curve equals the level 0.05

- With dependent d_t , replace $\sqrt{P}$ by $\sqrt{P/\Omega_d}$: the curves move to the right

Prerequisites for Today (7/20): structural VAR and cointegration

- ▣ **Recursive SVAR**: the reduced-form errors are a lower-triangular combination of structural shocks

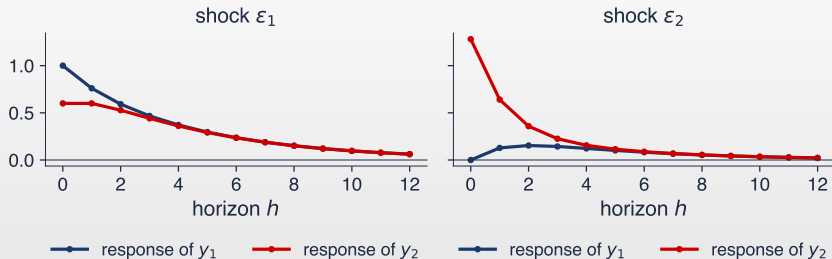
$$u_t = B_0 \varepsilon_t, \quad B_0 = \text{chol}(\Sigma_u), \quad \Theta_h = A^h B_0 \text{ (VAR(1))}$$

- ▶ Σ_u : covariance of u_t ; chol: Cholesky factor; A : VAR(1) coefficient matrix; Θ_h : responses at horizon h
- ▶ 2×2 : $b_{11} = \sqrt{\sigma_{11}}$, $b_{21} = \sigma_{21}/b_{11}$, $b_{22} = \sqrt{\sigma_{22} - b_{21}^2}$, with σ_{ij} the elements of Σ_u

- ▣ **VECM** without short-run terms: $\Delta y_t = \alpha \beta' y_{t-1} + \varepsilon_t$

- ▶ β : cointegrating vector; α : adjustment speeds; $z_t = \beta' y_t$: the equilibrium error, which follows an AR(1)
- ▶ $z_t = (1 + \beta' \alpha) z_{t-1} + \beta' \varepsilon_t$; half-life of a deviation: $\ln 0.5 / \ln(1 + \beta' \alpha)$ periods

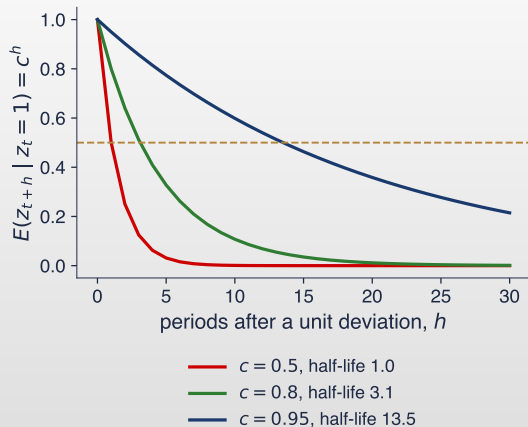
Prerequisites for Today (8/20): Cholesky impulse responses



□ Illustration: $A = \begin{pmatrix} 0.7 & 0.1 \\ 0.3 & 0.5 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 1 & 0.6 \\ 0.6 & 2 \end{pmatrix}$; responses $\Theta_h = A^h B_0$ for $h = 0, \dots, 12$

- ▶ B_0 : $b_{11} = 1$, $b_{21} = 0.60$, $b_{22} = 1.28$; $b_{12} = 0$, so y_1 does not respond on impact to ε_2 (right, $h = 0$)
- ▶ element (i, k) of Θ_h : the response of y_i after h periods to a one-standard-deviation shock ε_k ; stable VAR: all responses die out

Prerequisites for Today (9/20): the half-life of a deviation



- An AR(1) deviation $z_t = cz_{t-1} + \varepsilon_t$, $0 < c < 1$, decays on average as c^h

$$c^{h^*} = 0.5 \iff h^* = \frac{\ln 0.5}{\ln c}$$

- ▶ h^* : the half-life, in periods; here 1.0, 3.1 and 13.5 for $c = 0.5, 0.8, 0.95$
- The same formula serves the equilibrium error of a VECM ($c = 1 + \beta'\alpha$) and a GARCH volatility shock ($c = \alpha + \beta$)
- ▶ c close to 1: very slow return to equilibrium

Prerequisites for Today (10/20): the Kalman filter for a local level

- ▣ **Local level:** the observation is a random-walk level plus noise

$$y_t = \mu_t + \varepsilon_t, \quad \mu_{t+1} = \mu_t + \eta_t$$

- ▶ $\sigma_\varepsilon^2, \sigma_\eta^2$: variances of the observation and level noises; a, P : predicted level and its variance

- ▣ **One step:** prediction error variance, gain, update, next prediction

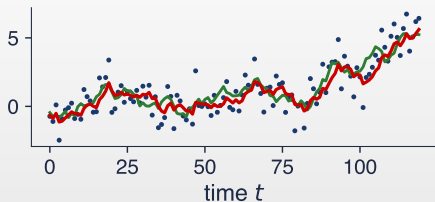
$$F = P + \sigma_\varepsilon^2, \quad K = P/F, \quad a_{t|t} = a + K(y - a), \quad P_{t|t} = P(1 - K), \quad P_{t+1} = P_{t|t} + \sigma_\eta^2$$

- ▶ $K \in (0, 1)$: the share of the surprise $y - a$ added to the level

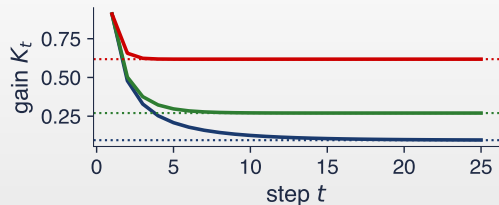
- ▣ **Steady state:** depends only on the signal-to-noise ratio $q = \sigma_\eta^2 / \sigma_\varepsilon^2$

$$\bar{P} = \sigma_\varepsilon^2 \frac{q + \sqrt{q^2 + 4q}}{2}$$

Prerequisites for Today (11/20): the Kalman filter at work



• y_t — level μ_t — filtered $a_{t|t}$



— $q = 0.01$ — $q = 0.1$ — $q = 1$

- Left: a simulated local level with $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 0.1$; the filtered level follows the true level and smooths the noise
- Right: the gain K_t from a vague start ($P_1 = 10$) converges in a few steps to $\bar{K} = \bar{P}/(\bar{P} + \sigma_\varepsilon^2)$ (dotted)
 - ▶ $\bar{K} = 0.095, 0.270, 0.618$ for $q = 0.01, 0.1, 1$: a noisier level (larger q) gives more weight to the new observation
 - ▶ with K constant the filter is exponential smoothing: $a_{t|t} = (1 - \bar{K})a_{t-1|t-1} + \bar{K}y_t$

Prerequisites for Today (12/20): Markov chains and GARCH

- ▣ **Two-state Markov chain:** p_{ij} is the probability of staying in regime i from one period to the next

$$\mathbb{E}D_i = \frac{1}{1 - p_{ii}}, \quad \pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}, \quad \lambda = p_{11} + p_{22} - 1$$

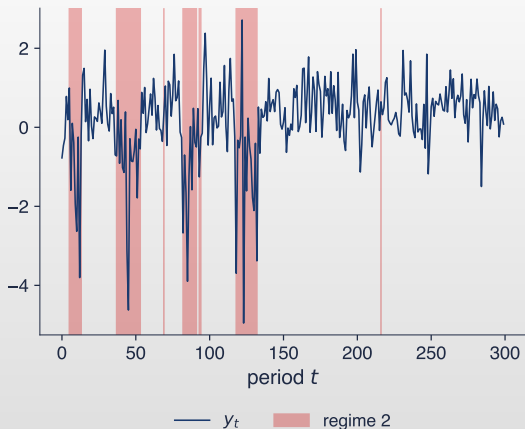
- ▶ $\mathbb{E}D_i$: expected duration of regime i ; π_1 : long-run (ergodic) share of time in regime 1; λ : persistence of the chain

- ▣ **GARCH(1,1):** today's variance from yesterday's squared shock and variance

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad \sigma^2 = \frac{\omega}{1 - \alpha - \beta}$$

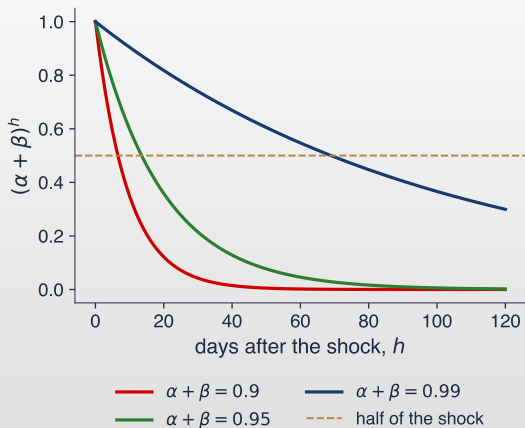
- ▶ $\omega > 0$, $\alpha, \beta \geq 0$; ε_{t-1} : yesterday's shock; σ^2 : unconditional variance (when $\alpha + \beta < 1$)
- ▶ persistence $\alpha + \beta$; half-life of a volatility shock $\ln 0.5 / \ln(\alpha + \beta)$ periods

Prerequisites for Today (13/20): a simulated two-state Markov chain



- $p_{11} = 0.97$ (calm regime), $p_{22} = 0.90$ (turbulent regime, shaded); y_t has a lower mean and a larger variance in regime 2
- Expected durations 33.3 and 10.0 periods; ergodic share of regime 1 $\pi_1 = 0.769$
 - ▶ in this sample of 300 periods regime 1 occupies 0.817 of the time
 - ▶ $\lambda = 0.87$: the regime of today predicts the regime of the next periods
- The duration of a regime is geometric:
 $\Pr(D_i = d) = p_{ii}^{d-1}(1 - p_{ii})$, $d = 1, 2, \dots$

Prerequisites for Today (14/20): the decay of a volatility shock



- GARCH(1,1) variance forecast h days ahead

$$\mathbb{E}_t \sigma_{t+h}^2 - \sigma^2 = (\alpha + \beta)^{h-1} (\sigma_{t+1}^2 - \sigma^2)$$

- ▶ σ^2 : the unconditional variance; a deviation of today's variance from σ^2 shrinks by the factor $\alpha + \beta$ each day
- Half-lives: 6.6, 13.5 and 69.0 days for $\alpha + \beta = 0.90$, 0.95, 0.99
- ▶ near 1 the half-life explodes: small changes in the persistence change it a lot

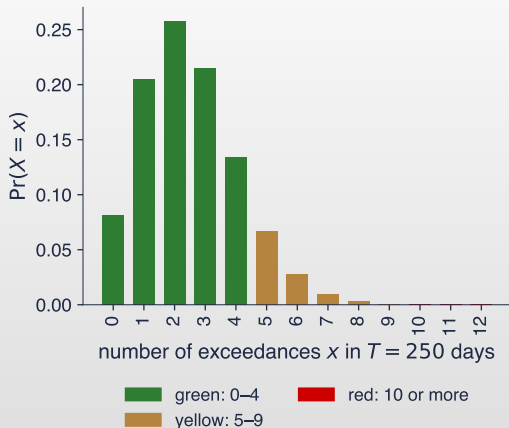
Prerequisites for Today (15/20): VaR backtests

- ▣ VaR 1%: $\text{VaR}_\alpha = -q_\alpha$, with q_α the α -quantile of the return and $\alpha = 0.01$
- ▣ **Kupiec test**: x exceedances in T days; is the exceedance rate equal to α ?

$$LR_{uc} = -2 \ln \frac{(1 - \alpha)^{T-x} \alpha^x}{(1 - \hat{\pi})^{T-x} \hat{\pi}^x} \sim \chi_1^2, \quad \hat{\pi} = x/T$$

- ▶ numerator: likelihood of the exceedances under the target rate α ; denominator: under the observed rate $\hat{\pi}$
- ▶ a large LR_{uc} (small p-value): too many or too few exceedances
- ▣ **Basel traffic light** for 250 days at 1%
 - ▶ green 0–4, yellow 5–9, red 10 or more exceedances

Prerequisites for Today (16/20): the binomial law of exceedances



- If the VaR is correct, the number of exceedances X in T days is Binomial(T, α)

$$\Pr(X = x) = \binom{T}{x} \alpha^x (1 - \alpha)^{T-x}$$

- ▶ $\binom{T}{x}$: the number of ways to place x exceedances among T days; $\mathbb{E}X = T\alpha = 2.5$ for $T = 250$, $\alpha = 0.01$
- The exact one-sided p-value of x_0 exceedances is the tail $\Pr(X \geq x_0) = \sum_{x=x_0}^T \Pr(X = x)$
- Basel zones for a correct model: green with probability 89.2%, yellow 10.8%, red 0.03%
 - ▶ a correct model falls in the yellow zone about one year in ten

Prerequisites for Today (17/20): conformal prediction

- ▣ **Split conformal** Romano, Patterson and Candès (2019): scores $S_1, \dots, S_n$ on a calibration set, e.g. absolute residuals

$$\hat{q} = S_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil$$

- ▶ $S_{(k)}$: the k -th smallest score; $\lceil \cdot \rceil$: rounding up; α : the allowed miss rate; the interval is $\hat{y} \pm \hat{q}$

- ▣ **Guarantee** under exchangeability (no ties)

$$1 - \alpha \leq \Pr(Y_{n+1} \in \hat{C}) \leq 1 - \alpha + \frac{1}{n+1}$$

- ▶ $\hat{C}$: the prediction interval; exchangeable: the joint law does not change when the points are reordered

- ▣ **ACI** Gibbs and Candès (2021) for dependent data: the working level α_t is updated after each observation

- ▶ $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$; γ : step; $\text{err}_t = 1$ after a miss: a miss widens the next interval

Prerequisites for Today (18/20): searches, placebos and the defence

- ▣ **Multiple tests:** m tests at level α , p-values p_j , sorted $p_{(1)} \leq \dots \leq p_{(m)}$
 - ▶ Bonferroni: reject when $p_j \leq \alpha/m$
 - ▶ Holm: reject while $p_{(i)} \leq \alpha/(m - i + 1)$, stop at the first failure; both control the FWER (probability of at least one false rejection)
 - ▶ max- t (Reality Check logic [White \(2000\)](#)): compare the best t with the bootstrap distribution of $\max_k t_k^*$, resampled jointly in blocks
- ▣ **Synthetic control placebo** [Abadie, Diamond and Hainmueller \(2015\)](#): rank of the treated unit among $J + 1$ units
 - ▶ p-value = $\#\{j : r_j \geq r_1\}/(J + 1)$; r_j : post/pre RMSPE ratio of unit j ; the smallest p-value is $1/(J + 1)$
- ▣ **Leakage:** any choice (selection, scaling, tuning) that uses data after the forecast origin [Kaufman et al. \(2012\)](#)
- ▣ At the defence: individual, three blocks of questions (code, results, chapters), no AI tools
 - ▶ a good answer: the number, its uncertainty, the assumption behind it, what would change it

Prerequisites for Today (19/20): a rule search in formulas

- ▣ **Moving-average rule** with window n : long the next day if the close is above its n -day average, otherwise cash

$$s_{n,t} = 1 \left\{ P_t > \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \right\}, \quad f_{n,t+1} = r_{t+1}(s_{n,t} - 1)$$

- ▶ P_t : the close; r_{t+1} : the next log return; $f_{n,t+1}$: the return of the rule minus that of buy-and-hold
- ▶ $t_n = \bar{f}_n / \text{se}_{\text{HAC}}(\bar{f}_n)$ for each of the K rules; the best rule has $t_{\max} = \max_n t_n$

- ▣ **Stationary bootstrap**: resample the days in blocks of random length, geometric with mean b , wrapping around the end

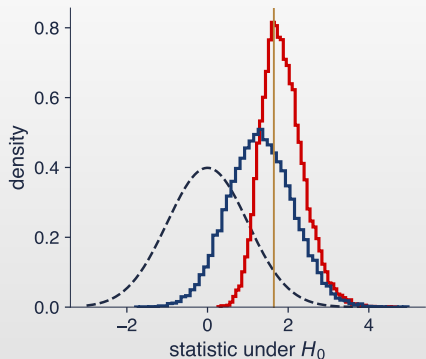
- ▶ the same days for all K rules: the correlation of the rules is kept; blocks keep the dependence over time

- ▣ **Max- t p-value** with B resamples of the centred differentials $f_{n,t} - \bar{f}_n$

$$t_{\max}^{*(b)} = \max_n \frac{\bar{f}_n^{*(b)} - \bar{f}_n}{\text{se}(\bar{f}_n)}, \quad p = \frac{1}{B} \sum_{b=1}^B 1 \{ t_{\max}^{*(b)} \geq t_{\max} \}$$

- ▶ the critical value is the 95% quantile of $t_{\max}^*$, not 1.645

Prerequisites for Today (20/20): the maximum of many t -statistics



--- one t , $N(0, 1)$ max of 20, correlation 0.5
 max of 20, correlation 0 — 1.645 (one test, 5%)

- Simulated: 20 statistics $N(0, 1)$ under H_0 , independent or with pairwise correlation 0.5
 - ▶ the maximum is far to the right of a single $N(0, 1)$
- 95% quantiles of the maximum: 2.79 (independent), 2.65 (correlation 0.5), against 1.645 for one test
 - ▶ with 20 independent tests at 5%, at least one rejection occurs with probability $1 - 0.95^{20} = 64\%$
 - ▶ the more correlated the rules, the closer the maximum stays to a single test

A1: how much does a mean know? (Chapter 0) [Solved]

Task

- A series is AR(1) with $\phi = 0.6$ and $\sigma^2 = 1$. A team reports $t = 2.5$ for its mean with i.i.d. standard errors.
- 1. Compute γ_0 and Ω .
- 2. By what factor is the i.i.d. standard error too small?
- 3. Compute the corrected t and state the conclusion at 5%.
- Report: two numbers; one number; one number and one sentence.

Solution

- 1. $\gamma_0 = 1/(1 - 0.36) = 1.5625$; $\Omega = 1/(1 - 0.6)^2 = 6.25$
- 2. $\Omega/\gamma_0 = (1 + \phi)/(1 - \phi) = 4$: the standard error is too small by a factor of $\sqrt{4} = 2$
- 3. $t = 2.5/2 = 1.25 < 1.96$: not significant
- Interpretation: at the defence, the first question about any t -statistic is "which standard error?"

A2: a Diebold–Mariano test by hand (Chapter 1) [Proposed]

Solution discussed in class

Task

- ▣ Squared-error differentials d_t (benchmark minus model), $h = 1$: 0.8, -0.2, 0.5, 1.1, 0.3, -0.4, 0.9, 0.6, 0.2, 0.4.
Model: [A1](#).
- ▣ 1. Compute $\bar{d}$, $\hat{\gamma}_0$ and $\hat{\gamma}_1$ (divide by P).
- ▣ 2. Compute $\hat{\Omega}$ with the Bartlett kernel and $L = 1$, then DM and HLN.
- ▣ 3. Compare with the critical value of t_9 and state the conclusion.
- ▣ Report: three numbers; three numbers; one sentence.

A3: Cholesky responses (Chapter 3) [Solved]

Task

- A VAR(1) $y_t = Ay_{t-1} + u_t$ with $A = \begin{pmatrix} 0.5 & 0 \\ 0.2 & 0.4 \end{pmatrix}$,
 $\Sigma_u = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$, recursive ordering.
- 1. Compute B_0 .
- 2. Compute $\Theta_1 = AB_0$ and $\Theta_2 = A^2B_0$.
- 3. What changes if the ordering is reversed?
- Report: a matrix; two matrices; one sentence.

Solution

- 1. $b_{11} = \sqrt{4} = 2.0$, $b_{21} = 2/2 = 1.0$, $b_{22} = \sqrt{5-1} = 2.0$
- 2. $\Theta_1 = \begin{pmatrix} 1.00 & 0.00 \\ 0.80 & 0.80 \end{pmatrix}$, $\Theta_2 = \begin{pmatrix} 0.50 & 0.00 \\ 0.52 & 0.32 \end{pmatrix}$
- 3. Reversed, the first variable responds on impact to the second shock and the second no longer to the first; with $\sigma_{12} \neq 0$ all responses change: the ordering is the identifying assumption
- Interpretation: a defence question is always "why this ordering?"

A4: the half-life of an equilibrium error (Chapter 4) [Proposed]

Solution discussed in class

Task

- A bivariate VECM with $\beta = (1, -1)'$ and $\alpha = (-0.2, 0.05)'$, no short-run terms. Model: A3.
- 1. Derive the AR(1) of $z_t = \beta' y_t$ and its coefficient.
- 2. Compute the half-life of a deviation.
- 3. Which variable is weakly exogenous if $\alpha_2 = 0$, and how is that tested?
- Report: one equation and one number; one number; one sentence.

A5: one step of the Kalman filter (Chapter 6) [Solved]

Task

- Local level with $\sigma_\varepsilon^2 = 4$, $\sigma_\eta^2 = 1$; predicted state $a = 10$, $P = 2$; observation $y = 13$.
- 1. Compute v , F and K .
- 2. Compute $a_{t|t}$, $P_{t|t}$ and P_{t+1} .
- 3. Compute the steady-state $\bar{P}$ and gain.
- Report: three numbers; three numbers; two numbers.

Solution

- 1. $v = 3$, $F = 2 + 4 = 6$, $K = 2/6 = 0.333$
- 2. $a_{t|t} = 10 + K \cdot 3 = 11$; $P_{t|t} = 2(1 - K) = 1.333$; $P_{t+1} = 2.333$
- 3. $q = 0.25$: $\bar{P} = 2.562$, $\bar{K} = \bar{P}/(\bar{P} + 4) = 0.390$
- Interpretation: the filter moves one third of the way towards the surprise; the signal-to-noise ratio q alone sets the long-run gain

A6: a two-state Markov chain (Chapter 7) [Proposed]

Solution discussed in class

Task

- ▣ Regime 1 (expansion) with $p_{11} = 0.95$, regime 2 (recession) with $p_{22} = 0.80$, monthly data. Model: A5.
- ▣ 1. Compute the expected durations.
- ▣ 2. Compute the ergodic probability of regime 1 and the persistence λ .
- ▣ 3. Why can a likelihood-ratio test of one regime against two not use χ^2_2 ?
- ▣ Report: two numbers; two numbers; one sentence.

A7: the Kupiec test for VaR 1% (Chapter 9) [Solved]

Task

- A VaR 1% model has $x = 6$ exceedances in $T = 250$ days.
- 1. Compute the hit rate and the expected number of exceedances.
- 2. Compute LR_{uc} and its p-value; compare with the exact binomial tail $\Pr(X \geq 6)$.
- 3. Give the Basel zone and the conclusion.
- Report: two numbers; three numbers; one sentence.

Solution

- 1. 2.4%; expected 2.5
- 2. $LR_{uc} = 3.56$, $p = 0.059$; exact one-sided binomial $p = 0.041$
- 3. Yellow zone; the asymptotic test does not reject at 5%, the exact tail does: with $\alpha = 1\%$ and 250 days, report both
- Interpretation: backtests on a few tail days have little power; one exceedance changes the verdict

A8: GARCH persistence (Chapter 8) [Proposed]

Solution discussed in class

Task

- ▣ Daily returns in %, GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$. Model: [A7](#).
- ▣ 1. Compute the persistence and the half-life of a volatility shock.
- ▣ 2. Compute the unconditional variance and the annualised volatility (250 days).
- ▣ 3. Why are Hessian standard errors of α misleading with fat-tailed returns?
- ▣ Report: two numbers; two numbers; one sentence.

A9: the split-conformal quantile (Chapter 13) [Solved]

Task

- ▣ Absolute residuals of 19 calibration points: 0.12, 0.85, 0.40, 1.30, 0.07, 0.66, 2.10, 0.95, 0.31, 1.75, 0.52, 0.22, 1.05, 0.48, 0.90, 1.42, 0.18, 0.73, 1.20; $\alpha = 0.1$.
- ▣ 1. Compute k and $\hat{q}$.
- ▣ 2. Give the coverage bounds of $\hat{y} \pm \hat{q}$.
- ▣ 3. Which assumption fails for daily returns, and what replaces it?
- ▣ Report: two numbers; two numbers; one sentence.

Solution

- ▣ 1. $k = \lceil 20 \cdot 0.9 \rceil = 18$: the 18th smallest score, $\hat{q} = 1.75$
- ▣ 2. Between 0.90 and $0.9 + 1/20 = 0.950$, marginally
- ▣ 3. Exchangeability fails with volatility clustering; ACI or weighted conformal restore long-run coverage, conditional coverage must be tested
- ▣ Interpretation: a guarantee is only as good as its assumption; say which one you rely on

A10: placebo p-values and Holm (Chapters 14 and 0) [Proposed]

Solution discussed in class

Task

- ▣ A synthetic control with 19 donors; the treated unit has the largest post/pre RMSPE ratio. The team tests five outcomes with $p = 0.010, 0.020, 0.040, 0.200, 0.030$. Model: [A9](#).
- ▣ 1. Compute the placebo p-value and the smallest attainable value.
- ▣ 2. How many outcomes are significant at 5% without correction, with Bonferroni and with Holm?
- ▣ 3. What should the pre-registration have said?
- ▣ Report: two numbers; three counts; one sentence.

B1: Meese and Rogoff for EUR/RON [Solved]

- **Question:** a team replicates [Meese and Rogoff \(1983\)](#): can an AR(1) beat the random walk for the monthly EUR/RON change? You are the examiner and the candidate at once
- **Inputs:** BNR reference rate, end of month, $100 \times \log$ change; rolling window of 60 months; 128 one-month forecasts from January 2016
- **Tasks**
 1. Produce the output: RMSE of both models, their ratio, DM–HLN and Clark–West.
 2. Defence question: why is Clark–West the right test here, and what does it test?
 3. Defence question: is the non-rejection evidence that the random walk is best? Use the power formula.
 4. Interpretation: write the one-sentence conclusion you would defend.
- **Report:** four numbers and two test statistics, two short answers, one sentence
- **Notebook:** section [B1](#)

B1: solution [Solved]



- RMSE: random walk 0.509, AR(1) 0.524; ratio 1.028; DM–HLN = 1.19 ($p = 0.235$); Clark–West = 1.11 (one-sided $p = 0.134$)
- Clark–West: the random walk is nested in the AR(1); under the null the estimated coefficients add noise to the larger model, which biases DM towards the random walk; CW corrects for it [Clark and West \(2007\)](#)
- Power: first-order autocorrelation of d_t -0.01 ; a gain of 0.2 standard deviations needs about 196 forecasts for 80% power, more than the 128 available
- Interpretation: “the Meese–Rogoff result holds for the leu over 2016–2026: the AR(1) does not beat the random walk, and our sample could only have detected a large gain”

B2: the term spread and growth [Proposed]

- **Question:** a team replicates [Estrella and Hardouvelis \(1991\)](#) and reports “the spread predicts US growth, $t = 4.7$ ”. Defend or correct it
- **Inputs:** FRED real GDP, 10-year and 3-month yields (quarterly averages), 1962–2025; $y_t = (400/4) \ln(\text{GDP}_{t+4}/\text{GDP}_t)$ on the spread; model: [B1](#)
- **Tasks**
 1. Estimate the slope with classical, Hansen–Hodrick-type (3 lags) and Newey–West standard errors.
 2. Defence question: why must the errors be at least MA(3)? Report the residual autocorrelation.
 3. Defence question: is the relation stable? Estimate it before 1989 and from 1989.
 4. Interpretation: rewrite the team’s sentence so that it matches the evidence.
- **Report:** a slope, three standard errors and t -statistics, one autocorrelation, two subsample slopes, one sentence
- **Notebook:** section [B2](#)

B3: a leakage audit [Solved]

- **Question:** a team selects the 5 best of 100 candidate predictors by their correlation with the target over the whole sample, then reports a positive out-of-sample R^2 on the last 60 periods. Is the gain real?
- **Inputs:** a simulation with pure-noise predictors and target, $T = 240$, 1 000 replications; model: [B1](#)
- **Tasks**
 1. Run the team's protocol and the honest protocol (selection on the training sample only).
 2. Report the mean out-of-sample R^2 and the share of positive values for both.
 3. Defence question: which line of code leaks, and how is it fixed?
 4. Interpretation: what would you require before believing a reported R^2 of 10%?
- **Report:** four numbers, one line of code, one sentence
- **Notebook:** section [B3](#)

B3: solution [Solved]

- ▣ Team protocol: mean $R^2 = 5.9\%$, positive in 81% of replications (90th percentile 14.4%)
- ▣ Honest protocol: mean $R^2 = -13.3\%$, positive in 8%: the predictors are noise
- ▣ The leak: the correlations are computed on `rows = slice(0, T)`; the fix is `slice(0, n_train)`, and in a rolling design the selection is repeated at every origin
- ▣ Interpretation: a R^2 of 10% is within the range that the leak alone produces; ask for the selection code, a placebo with shuffled targets and a test period never touched

B4: a rule search on the BET [Proposed]

- **Question:** a team reports “the 5-day moving-average rule beats buy-and-hold on the BET, $p = 0.006$ ”. It tried 20 rules. Defend or correct it
- **Inputs:** BET daily log returns, 2001–2012 (2 993 days); rules “long if the close is above its n -day average”, $n = 5, 10, \dots, 100$; HAC t of the mean excess return; model: [B3](#)
- **Tasks**
 1. Compute the t -statistic of every rule and the naive p -value of the best.
 2. Compute the max- t p -value with the stationary bootstrap (mean block 10 days, 999 resamples).
 3. Defence question: why does the bootstrap resample the 20 series jointly, in blocks?
 4. Interpretation: what would you add before calling it a profitable rule?
- **Report:** 20 statistics, two p -values, one critical value, one sentence
- **Notebook:** section [B4](#)

C1: the pre-registration of your project [Proposed]

- **Question:** is your planned evaluation able to answer your question?
- **Inputs:** your team's landmark paper and data; the power formula of today's formula sheet; model: [B1](#)
- **Tasks**
 1. Write the seven sections of the pre-registration in at most one page: question and hypotheses, paper to replicate, data, models, evaluation, robustness, team.
 2. Write the rule that decides "replicated", "partly" or "not replicated".
 3. Compute the evaluation length needed for 80% power at the gain reported by the paper, with ρ estimated on your training data.
 4. Interpretation: if the needed length exceeds your data, how do you change the question?
- **Report:** one page, one rule, one number, one sentence
- **Notebook:** section [C1](#)

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant how to prepare the replication project and its defence. The answer:
- ▣ (a) Use the Diebold-Mariano test to compare your AR(1) with the random walk; it is valid for nested models.
- ▣ (b) With 19 donors, a placebo p-value of 0.01 is achievable if the effect is large.
- ▣ (c) If your result is not significant, the random walk is the true model.
- ▣ (d) For an AR(1) with $\phi = 0.6$, i.i.d. standard errors of the mean are too small by a factor of 2.
- ▣ (e) Standardise the predictors once on the full sample before the walk-forward loop: it saves time and changes nothing.
- ▣ (f) Holm rejects more hypotheses than Bonferroni and controls the same family-wise error rate.
- ▣ (g) You may use the assistant during the oral defence if you declare it in AI_USE.md.
- ▣ **Tasks**
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ Every number at the defence comes with its standard error, and the standard error with its assumption
- ▣ Nested models need Clark–West; a non-rejection needs a power calculation
- ▣ Overlapping targets and persistence enlarge standard errors; stability over subsamples is part of the evidence
- ▣ Selection on the whole sample creates out-of-sample gains from noise; a search needs a search-adjusted p-value
- ▣ Write the pre-registration and the verdict rule before the first estimate

After the seminar

- Lecture 15 develops each topic of today
 - ▶ the course map, one recap per chapter, the research workflow, the replication scoreboard
 - ▶ the failure modes, the oral defence and the AI policy
- Try the [Proposed] tasks in the notebook
- C1 is the first draft of your team's proposal and pre-registration
- Reading: [Clemens \(2017\)](#); [Nosek et al. \(2018\)](#); [Diebold \(2015\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- ▣ Abadie, A., Diamond, A., & Hainmueller, J. (2015). Comparative politics and the synthetic control method. *American Journal of Political Science*, 59(2), 495–510. doi:10.1111/ajps.12116
- ▣ Clark, T. E., & West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311. doi:10.1016/j.jeconom.2006.05.023
- ▣ Clemens, M. A. (2017). The meaning of failed replications: A review and proposal. *Journal of Economic Surveys*, 31(1), 326–342. doi:10.1111/joes.12139
- ▣ Diebold, F. X. (2015). Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of Diebold–Mariano tests. *Journal of Business & Economic Statistics*, 33(1), 1–9. doi:10.1080/07350015.2014.983236
- ▣ Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- ▣ Estrella, A., & Hardouvelis, G. A. (1991). The term structure as a predictor of real economic activity. *The Journal of Finance*, 46(2), 555–576. doi:10.1111/j.1540-6261.1991.tb02674.x
- ▣ Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems 34*; arXiv:2106.00170. [arxiv.org](https://arxiv.org/abs/2106.00170)
- ▣ Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. doi:10.1016/S0169-2070(96)00719-4
- ▣ Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), 15. doi:10.1145/2382577.2382579
- ▣ Meese, R. A., & Rogoff, K. (1983). Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics*, 14(1–2), 3–24. doi:10.1016/0022-1996(83)90017-X
- ▣ Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. doi:10.2307/1913610
- ▣ Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606. doi:10.1073/pnas.1708274114
- ▣ Romano, Y., Patterson, E., & Candès, E. (2019). Conformalized quantile regression. *Advances in Neural Information Processing Systems 32*; arXiv:1905.03222. [arxiv.org](https://arxiv.org/abs/1905.03222)
- ▣ White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126. doi:10.1111/1468-0262.00152